TIANYU LU

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EDUCATION

Texas Academy of Mathematics and Science

University of North Texas | *Denton, TX* Aug. 2025-May 2027 (Expected)
Residential early-college program; university coursework in Grades 11-12.
Current GPA: 3.889/4.00

Jasper High School | *Plano, TX* Aug. 2023-May 2025
Completed Grades 9-10 before entering TAMS.

ACADEMIC AND TECHNICAL ACTIVITIES

Student Researcher | *University of North Texas, Prof. Qing Yang's Lab* 11-12
12 hrs/wk, 44 wks/yr

- Developed an attention-enhanced reinforcement-learning model to assign ground robots to tasks identified by a simulated aerial drone, coordinating which robot should respond to each task.
- Implemented and evaluated a ROS 2, Gazebo, and Open-RMF simulation with three ground robots and 20 tasks in an airport-terminal map; explored adapting the coordination problem to search and rescue.
- Compared six scheduling methods. In the tested simulation batch, the model reduced mean completion time by 16% versus standard PPO and produced the lowest queue variance; examined trade-offs because a greedy method performed better on local waiting and travel measures.
- Prepared an eight-page first-author manuscript and submitted it on August 1, 2026 to the IEEE Annual Congress on Artificial Intelligence of Things; last reported status: under review.

UAV Builder and Racing Drone Hardware Lead | *Independent Project and EkDrone Team* 11-12
5 hrs/wk, 48 wks/yr

- Independently built a sensor-equipped drone platform for hands-on exploration of autonomous navigation and future real-world data collection; integrated a depth camera and onboard computing hardware.
- Co-built a racing drone with three peers, taking responsibility for hardware integration, component compatibility, receiver setup, and remote-control tuning; adjusted flight parameters through hands-on testing.
- Ran pre-flight risk checks and piloted the racing drone's maiden flight after the team verified the system; enjoyed experimenting with how hardware choices and control settings changed the flying experience.

Research Assistant | *UNT Autonomous-Driving Perception Project* 11-12
5 hrs/wk, 44 wks/yr

- Helped build an MMDetection-based pipeline combining Transformer networks and occupancy modeling to locate nearby vehicles from camera images; tested the stability and complexity of six-camera versus single-camera inputs.
- Tackled challenges in perception, enabling vehicles to accurately interpret their surroundings through visual data
- Contributed to a subsequent GiT multi-task vision study covering object detection, image segmentation, captioning, and locating objects from text; ran experiments, analyzed training and evaluation logs, and prepared visualizations.
- Helped investigate task-conditioned routing between two expert pathways. Analyzed why separating feature representations did not consistently improve accuracy and how related visual tasks could share computational resources.

Team Researcher | *UT Dallas STEM Bridge Summer Research Camp* 10

15 hrs/wk, 6 wks/yr

- Worked with four peers under Dr. Jie Zhang at UT Dallas on machine-learning-based monitoring of wind-turbine generator windings using electrical signals.
- Learned Python data visualization and machine-learning methods; helped evaluate an LSTM model for classifying generator faults into mild, medium, and severe levels.
- Compared clean and noisy signal results. The team poster reported 100% validation accuracy on clean data and 46.0% at a signal-to-noise ratio of 10, highlighting limits under noise rather than demonstrated field reliability.
- Presented the team research poster explaining the monitoring approach and model evaluation; the team received the program's Best Science Education Award.

Team Researcher | *Wing Washout Aerodynamics Study* 10

25 hrs/wk, 10 wks/yr

- Motivated by flight safety, studied whether twisting a wing could delay wing-tip stall without imposing excessive efficiency costs; used a simplified Boeing 737 wing model to examine that trade-off.
- Built and meshed the model in ANSYS Fluent and compared 0°/3°/5°wings across angles of attack; interpreted simulated flow separation and lift-to-drag performance rather than claiming a flight-tested design.
- Troubleshot negative-volume mesh errors through local refinement and boundary-layer settings; used the SST k-omega turbulence model to better represent near-wall flow and separation.
- Presented the study at the school science fair and documented the need for numerical checks and comparison with reference or experimental data before drawing real-aircraft safety conclusions.

Engineering Project Team Member | *School Marble-Sorter Project* 10

8 hrs/wk, 4 wks/yr

- Built a system to distinguish and route wood, glass, and metal marbles using observable material and color differences.
- Diagnosed unreliable glass-marble detection caused by similar colors and added blue tape at the exit so the sensor could register glass marbles consistently as they passed.

LEADERSHIP, COMMUNICATION & SERVICE

Founder, Host and Producer | Ximalaya Technical Podcasts **2018-Present**

2 hrs/wk, 30 wks/yr

- Began publishing recitations and English dubbing in 2018, building long-term practice in vocal delivery, pacing, recording, and revision.
- Relaunched the channel in late 2025 with 23 original episodes: 15 episodes of Engineering Insights and 8 episodes of Self-Driving Beginner's Journal.
- Researched documented engineering accidents and explained the design and software decisions behind them in accessible Chinese-language scripts, asking how systems fail and what could prevent similar failures.
- Produced a self-driving learning journal on perception and decision-making, using explanations aloud to test understanding and correct misconceptions; the combined programs have received 16.8k plays.

Founder and President | *Jasper High School Physics Club* 10

2 hrs/wk, 16 wks/yr

- Founded the school's first physics club and recruited more than 60 members.
- Planned recurring workshops on college-level physics topics and demonstrations, prepared explanations and practice problems, and coordinated member participation.
- Formed and coached a Physics Bowl team that earned second place in Texas; transferred club responsibilities to a successor when leaving Jasper for TAMS.

Physics Competition Committee Head | *Mu Alpha Theta, TAMS* 2026-Present
2 hrs/wk, 16 wks/yr

- Leads weekly mechanics sessions and works through past F=ma problems with students preparing for physics competitions.
- Assembled and organized a reusable problem bank from prior examinations to support structured practice and discussion.

Volunteer Mathematics Tutor | *Stephens Elementary School* 11-12
1.5 hrs/wk, 10 wks/yr

- Tutors 20 fourth- and fifth-grade students in mathematics through weekly small-group instruction and problem-solving practice.
- Organized MathFun, a game-based session designed to draw students into mathematical reasoning through accessible challenges.

Fundraising and Service Volunteer | *Alliance of Youth Leaders in the United States* 9-10
1 hr/wk, 16 wks/yr

- Operated concession stands at community parks and personally raised approximately $300 for programs serving children with disabilities.
- Packed donated books for communities in Africa and prepared meal packages through Feed My Starving Children.

ATHLETICS

Point Guard and Independent Strength Trainee | *TAMS Basketball Team* 2023-Present
5 hrs/wk, 40 wks/yr

- Played as point guard in the UNT Open Basketball Tournament and participated in regular team practices and games.
- Maintains a near-daily independent routine combining strength training and running, continuing a habit established in middle school.

Player | *TAMS Table Tennis Team* 2025-Present
2 hrs/wk, 16 wks/yr

- Competes weekly in club singles and doubles matches; practices forehand and backhand technique, footwork, and match strategy.

Bridge Player 2014-2023
2 hrs/wk, 40 wks/yr

- Competed in tournament bridge from Grade 1 through Grade 9 (2014–2023).
- Learned bidding systems, card-play strategy & partnership communication through regular training and local tournaments.

AWARDS & HONORS

- UNT Undergraduate Research Fellowship (Grade 11-12)
- USA Mathematical Olympiad (USAMO) Qualifier; AIME best score: 12/15 (Grades 11)
- AMC 10 Distinguished Honor Roll, top 1%; AMC 12 Distinction, top 5% (Grades 10-11)
- USA Physics Olympiad (USAPhO) Qualifier via the F=ma Exam (Grade 10)
- Physics Bowl, 2nd-Place Team in Texas, Region 11, Division 1 (Grade 10)
- Plano ISD Superintendent's Scholar Award, district-wide (Grade 10)
- Best Science Education Award, Texas Association of Science and Technology STEM Bridge Summer Research Camp (Grade 10)

TECHNICAL SKILLS

Autonomous systems and machine learning: Python-based reinforcement learning; ROS 2; Gazebo; Open-RMF; MMDetection; multi-task vision experiment analysis.
Drone hardware and controls: Component integration; receiver and remote-control setup; flight tuning; pre-flight checks; 3D-printed components.
Aerodynamic simulation: ANSYS Fluent modeling and meshing; boundary-layer settings; comparison of wing twist and angle-of-attack cases.
Communication: Research writing; technical podcast scripting, recording, and production.
Programming languages: Python, Java, C++, JavaScript, HTML, CSS